# 5.5.6

Minimal Sufficient Statistics

## Concept

A **minimal sufficient statistic** is a sufficient statistic that does not include any extraneous information needed to describe the distribution from which it comes.

## Definition 5.6.1

Let $X_1, X_2, \ldots, X_n$ have joint pdf $f(\vec{x}; \theta)$. A statistic $S$ is *minimal sufficient* for this model if

- $S$ is sufficient, and
- $S$ is a function of every other sufficient statistic for the model.

## Bahadur's Theorem

If $S$ is complete and sufficient, then $S$ is minimal sufficient (assuming one exists).
This is true for almost any parameteric model.

## Example 5.6.1

Let $X_1, X_2, \ldots, X_n$ be a random sample from the $N(0, \sigma^2)$ distribution. Determine the minimal sufficient statistic for the distribution.

## Example 5.6.1 (cont)

## Example 5.6.1 (cont)

## Theorem 5.6.1

Suppose that $X_1, X_2, \ldots, X_n$ have joint pdf $f(\vec{x}; \theta)$.
If a statistic $T = t(\bar{X})$ has the following property:

$$\frac{f(\vec{x}; \theta)}{f(\vec{y}; \theta)} \text{ is } \theta\text{-free} \quad \Longleftrightarrow \quad t(\vec{x}) = t(\vec{y}),$$

then $T$ is a minimal sufficient statistic.

## Example 5.6.2

Let $X_1, X_2, \dots, X_n \overset{i.i.d.}{\sim} N(\theta, \theta^2)$. Use Theorem 5.6.1 to determine the minimal sufficient statistics for $(\mu, \sigma^2)$.

## Example 5.6.2 (cont)

## Example 5.6.2 (cont)

## Example 5.6.2 (cont)

# 5.5.7

Ancillary Statistics and Basu's Theorem

## Definition 5.7.1

An **ancillary statistic** is a statistic whose distribution does not depend on any unknown parameters.

E.g., if $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$, with $\sigma^2$ known, then $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$ is an ancillary statistic.

## More context

An ancillary statistic is the antithesis to a sufficient statistic.
A sufficient statistic contains all the information we need about the parameters, while an ancillary statistic contains no information about the parameters.

## 5.7.1

Location Parameters and Location-Invariant Statistics

## Definition 5.7.2

$\theta$ is said to be a location parameter for a distribution if the pdf $f(x;\theta)$ can be written as
$f(x;\theta) = f_0(x-\theta)$
for some function $f_0$ does not depend on $\theta$.
E.g., $\mu$ is a location parameter for the $N(\mu,\sigma^2)$ distribution.

## Definition 5.7.3

A statistic $T = t(\bar{X})$ is a location-invariant statistic if
$t(x_1+c, x_2+c, \ldots, x_n+c) = t(x_1, x_2, \ldots, x_n)$
for all $c \in \mathbb{R}$.

## Example 5.7.1

Let $X_1, X_2, \ldots, X_n$ be a random sample from any distribution.
Let $Y_i = X_i + c$ for $i = 1, 2, \ldots, n$.

## Theorem 5.7.1

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} f(x;\theta)$.
If $\theta$ is a location parameter and $T$ is a location-invariant statistic, then $T$ is ancillary.

## Example 5.7.2

Suppose that $X_1, X_2, \ldots, X_n$ is a random sample from the shifted exponential distribution with rate $1$ with pdf
$f(x;\theta) = e^{-(x-\theta)} I_{(0,\infty)}(x)$.
Determine an ancillary statistic.

## Example 5.7.2 (cont)

### Example 5.7.2 (cont)

### Example 5.7.2 (cont)

## 5.7.2

Scale Parameters and Scale-Invariant Statistics

### Definition 5.7.4

$\theta$ is said to be a scale parameter for a distribution if the pdf $f(x;\theta)$ can be written as

$$f(x;\theta) = \frac{1}{\theta} f_0(x/\theta)$$

for some function $f_0$ that does not depend on $\theta$.

$\theta$ is an inverse scale parameter if the pdf $f(x;\theta)$ can be written as

$$f(x;\theta) = \theta f_0(\theta x)$$

for some $f_0$.

E.g., $\lambda$ is an inverse scale parameter for the $\mathrm{Exponential}(\lambda)$ distribution.

## Definition 5.7.5

A statistic $T = t(\bar{X})$ is a scale-invariant statistic if
$t(cx_1, cx_2, \ldots, cx_n) = t(x_1, x_2, \ldots, x_n)$
for all $c > 0$.

## Theorem 5.7.2

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with pdf $f(x;\theta)$. If $\theta$ is a scale (or inverse scale) parameter and $T$ is a scale-invariant statistic, then $T$ is ancillary.

## Example 5.7.3

Suppose that $X_1, X_2, \ldots, X_n$ is a random sample from the $\mathrm{Gamma}(2, \beta)$ distribution. Determine an ancillary statistic.

## Example 5.7.3 (cont)

## Example 5.7.3 (cont)

## 5.7.3

Basu’s Theorem

## Theorem 5.7.3

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} f(x; \theta)$.
Suppose that $S$ is sufficient and $T$ is any statistic that is independent of $S$. Then $T$ is ancillary.

## Basu's Theorem

Let $X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} f(x; \theta)$. If $S$ is complete and sufficient and $T$ is ancillary, then $S$ and $T$ are independent.

## Example 5.7.4

Suppose that
$X_1, X_2, \ldots, X_n \overset{i.i.d.}{\sim} N(\mu, \sigma^2)$,
where $\sigma^2 > 0$ is known.
Show that the sample mean, $\bar{X}$, is independent of the sample range $X_{(n)} - X_{(1)}$.

## Example 5.7.4 (cont)

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